

Challenge (Habanero)

- Q1.** If $f(x) = x^2$, which function represents a vertical shift of $f(x)$ by 5 units up?
- (A) $f(x) = x^2 + 5$
(B) $f(x) = x^2 - 5$
(C) $f(x) = (x + 5)^2$
(D) $f(x) = (x - 5)^2$
(E) $f(x) = 5x^2$
- Q2.** A park ranger is tracking the population of deer in a forest. The population P after t years is given by $P(t) = 100(1.05)^t$. If the population is reflected over the t -axis, the new function is
- (A) $P(t) = -100(1.05)^t$
(B) $P(t) = 100(1.05)^{-t}$
(C) $P(t) = 100(0.95)^t$
(D) $P(t) = -100(0.95)^t$
(E) $P(t) = 100(1.05)^t$
- Q3.** A hiker is ascending a mountain with an elevation E given by $E(x) = 2000 + 50x$, where x is the horizontal distance. If the elevation is shifted 10 units to the left, the new function is
- (A) $E(x) = 2000 + 50(x + 10)$
(B) $E(x) = 2000 + 50(x - 10)$
(C) $E(x) = 2010 + 50x$
(D) $E(x) = 1990 + 50x$
(E) $E(x) = 2000 + 50x$
- Q4.** The fuel consumption F of a car is given by $F(x) = 20 + 0.5x$, where x is the distance traveled. If the fuel consumption is compressed horizontally by a factor of 2, the new function is
- (A) $F(x) = 20 + 0.5(2x)$
(B) $F(x) = 20 + 0.5(x/2)$
(C) $F(x) = 20 + x$
(D) $F(x) = 20 + 0.25x$
(E) $F(x) = 20 + 0.5x$
- Q5.** The temperature T of a lake is given by $T(x) = 50 - 2x$, where x is the depth below the surface. If the temperature is reflected over the x -axis, the new function is
- (A) $T(x) = 50 + 2x$
(B) $T(x) = 50 - 2x$
(C) $T(x) = -50 + 2x$
(D) $T(x) = -50 - 2x$
(E) $T(x) = 50 + 2x$
- Q6.** A bird's altitude A is given by $A(t) = 100 \cdot 2^{0.1t}$, where t is time in minutes. If the altitude is stretched vertically by a factor of 3, the new function is
- (A) $A(t) = 300 \cdot 2^{0.1t}$
(B) $A(t) = 100 \cdot 2^{0.3t}$
(C) $A(t) = 100 \cdot 2^{0.1t}$
(D) $A(t) = 100 \cdot 3^{0.1t}$
(E) $A(t) = 300 \cdot 3^{0.1t}$
- Q7.** A river's width W is given by $W(x) = 10 + 2x$, where x is the distance from the source. If the width is shifted 5 units down, the new function is
- (A) $W(x) = 5 + 2x$
(B) $W(x) = 10 + 2x$
(C) $W(x) = 15 + 2x$
(D) $W(x) = 10 - 2x$
(E) $W(x) = 5 - 2x$
- Q8.** A bike's speed S is given by $S(t) = 20 + 5t$, where t is time in hours. If the speed is reflected over the y -axis, the new function is
- (A) $S(t) = -20 - 5t$
(B) $S(t) = -20 + 5t$
(C) $S(t) = 20 - 5t$
(D) $S(t) = 20 + 5t$
(E) $S(t) = -20 - 5t$

Open Ended Questions (FRQ)

- Q9.** The population P of a species of fish is given by $P(t) = 500(1.2)^t$, where t is time in years. If the population is compressed horizontally by a factor of 2 and then reflected over the t -axis, find the new function and graph it. Show all work.
- Q10.** A hiker is descending a mountain with an elevation E given by $E(x) = 3000 - 100x$, where x is the horizontal distance. If the elevation is shifted 20 units to the right and then stretched vertically by a factor of 2, find the new function and explain its transformations. Show all work.

Chef's Tips

- Tip 1: Remind students to read each question carefully and identify the type of transformation.
- Tip 2: Encourage students to use graphing calculators or software to visualize the transformations.
- Tip 3: Emphasize the importance of showing all work and explaining their reasoning for the free-response questions.